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Inventor(s)	Shoup; David Loren et al.

Method to improve performance of a wireless data connection

Abstract

A method to improve network performance is disclosed. A preferred embodiment of the method involves active data management across multiple connections running processes that enable cohesive, robust and secure data communications across these multiple connections. This active data management method creates a meta network (network about a network) that significantly increases the quality and reliability of the overall communications network and can be applied with or without requiring full custody across the complete connection.

Inventors: **Shoup; David Loren (Huntington Beach, CA), Warner; Darren (Vancouver, WA)**

Applicant: **AIRBRIJ Technology Inc.** (Reno, NV)

Family ID: **1000008821069**

Assignee: **AIRBRIJ TECHNOLOGY INC. (Reno, NV)**

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Primary Examiner: Ngo; Ricky Q

Assistant Examiner: Jahangir; Kabir U

Attorney, Agent or Firm: Eastman IP

Background/Summary

RELATED APPLICATION (1) This application claims priority to U.S. Provisional Patent Application Ser. No. 63/136,032 for a “Method to improve performance of a wireless data connection,” filed Jan. 11, 2021, and currently co-pending, the entirety of which is incorporated herein by reference.

FIELD OF THE INVENTION
(1) The present invention pertains generally to transmission of data. The invention is particularly, but not exclusively, useful as a system and method for improving the reliability and throughput of wired and wireless data transfer in conditions of poor signal quality.
BACKGROUND OF THE INVENTION
(2) As the current utilization of the public Internet evolves, a number of limitations related to standard networking protocols have come to light. These limitations can be overcome utilizing the available networking standards (e.g., TCP/IP) in some cases, but in other cases, the available networking standards create issues, especially in situations where the connection medium is not sufficiently robust (e.g., wireless communications, existing suburban and rural wireline telecommunications networks). As utilization requirements change, in many situations, the existing telecommunications networks are insufficient to provide reliable and secure services, creating significant issues in the areas of performance, management and cybersecurity. In the case of wide-

area networks, these problems can be solved using robust software defined networking methods, however, they require a significant amount of computing power at their endpoints, also referred to as nodes (the active components that manage and route telecommunications traffic) and consume a high amount of bandwidth communicating control and management data (the “Control Plane”), which is not an issue for wide area networking, where the bandwidth is sufficient to absorb this overhead. In the case of consumer networks, these methods do not work most of the time because of the lack of expensive equipment at the endpoints combined with the constraint of bandwidth that is typical across the wired and wireless networks in most locales.

(3) With respect to wireless data transmission in general, as the distance increases between two wireless devices, the data performance also decreases due to a number of factors including, but not limited to, decreasing signal strength, radio interference, congestion, data error rates, and data management overhead.

(4) In view of the above, it would be advantageous to provide a method of active data management utilizing extremely efficient meta-networking methods that augments and enable management methods outside the available networking standards including novel methods of introducing efficiency, control and security across any type of telecommunications network.

(5) In the case of wireless connections, it would be further advantageous to provide a method of improving data connectivity that utilizes a number of different methods to improve the quality, speed, and reliability of data transmitted over a wireless connection.

SUMMARY OF THE INVENTION

(6) Embodiments of the disclosed active meta networking system and method can be applied across a number of applications.

(7) In a first exemplary preferred embodiment, as the distance of a wireless connection increases, it is subjected to degradation and the data performance also decreases due to a number of factors including, but not limited to, decreasing signal strength, radio interference, congestion, data error rates and data management overhead. The embodiment utilizes a number of different methods to improve the quality, speed and reliability of data transmitted over a wireless or wired connection solving the aforementioned problem. By controlling one or both ends of a data transmission, various network management methods can be combined to significantly improve the reliability and throughput of data, especially in locations where wireless connection quality varies.

(8) In this application, this embodiment is an improvement on what currently exists based on the methods of connectivity. Current methods of extending the range and quality of wireless connectivity generally utilize antenna technology to increase the range of the device. They do not address the issues introduced by extreme distance, interference or congestion. Without the ability to actively manage the data communication process, these devices cannot overcome the challenges that are introduced by poor connectivity of any kind.

(9) Preferred embodiments of the disclosed method of active data management run processes that enable cohesive data communications across multiple wireless connections at once across multiple wireless terminal devices. This eliminates a single point of failure and allows the processes to manage data across these connections in an orchestrated manner that improves reliability and throughput.

(10) In a second preferred embodiment, when a network is created with multiple remote nodes utilizing consumer wired connections, like that of an employee working from home, the connection quality and security is based on a number of variable factors related to the inherent design of these networks and consumer premises equipment (CPE). As local traffic increases, the network inherently degrades due to congestion. This is because consumer networks are not architected in a manner to deliver consistent quality of service or redundancy. CPE is also subject to potential mismanagement and cybersecurity breaches simply due to their implementation and underlying technology. By implementing edge services on an appropriate CPE connected to a cloud service, it is possible to create multiple paths of communication over both wired and wireless connections,

using them in tandem to mitigate the issues discussed above. It is also possible to overcome the cybersecurity issues in remote nodes using the same edge services

(11) In this application, this embodiment is an improvement on what currently exists based on the methods of the management of communications. Current methods of managing the quality and security of communications do not address the inherent bandwidth and architectural challenges of most current urban and rural wired and wireless networks. Without the ability to actively manage the data communication process across multiple connections, these CPE cannot overcome the challenges that are introduced when usage exceeds the capabilities of these networks.

(12) The disclosed embodiments of the method of active data monitoring and management running processes allow cohesive data traffic detection and communication management across multiple wired and wireless connections at once. This allows the aforementioned edge service to be directed to route specific data from devices connected to the CPE across different communications networks in a cohesive and seamless fashion to improve the overall quality of the network from the connected device's point of view. It also allows the cloud service to detect anomalous behavior on the CPE to be used to secure potential security breaches on that CPE and command the edge service to interrupt communications, thereby securing the network from a potential intrusion.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The novel features of this invention, as well as the invention itself, both as to its structure and its operation, will be best understood from the accompanying drawings, taken in conjunction with the accompanying description, in which similar reference characters refer to similar parts, and in which:

(2) FIG. **1** illustrates components of a preferred embodiment of a management platform operable to improve performance of wireless and wired networks;

(3) FIG. **2** illustrates the creation of peer-to-peer networks to create additional communication paths for devices on a network;

(4) FIG. **3** illustrates the use of the platform of FIG. **1** to actively manage both ends of a data transmission process maintaining full custody of the data before transmission over the Internet;

(5) FIG. **4** illustrates the use of the platform of FIG. **1** to actively manage one end of a data transmission process before transmission over the Internet;

(6) FIG. **5** illustrates the use of components to actively manage both ends of a data transmission process involving a wireless network; and

(7) FIG. **6** illustrates a preferred embodiment of a method to improve performance of wireless and wired networks.

DETAILED DESCRIPTION

(8) Referring initially to FIG. **1**, a preferred embodiment of a management platform for improving performance of wireless and wired networks is illustrated and generally designated **100**. Platform **100** has three components, including cloud service **110**, edge service **112**, and end-user interface **114**.

(9) Cloud service **110** is the central management and control process combined with methods to optimize data communication with one or more wired and/or wireless terminal devices **116**—also referred to herein as user devices—running the edge service **112**. For example, terminal devices **166** may include one or more of computer **116**A, tablet computing device **116**B, and mobile telephone **116**C.

(10) Edge service **112** operates on the wired and/or wireless terminal services **116**. Edge service **112** is focused on communications, interaction, telemetry, optimization, API management, and wired and/or wireless terminal device **116** management. Edge service **112** is capable of executing

across multiple wired and/or wireless devices **115** seamlessly, creating a single virtualized edge service **112** that acts as the remote node end of the data management process. This edge service **112** running across all devices **115** provides, in conjunction with cloud service **110**, authentication and security, data management, Internet of Things (IoT) management, and manages application and data service APIs. Because of their close association with edge service **112**, devices **115** may be referred to herein as “service devices.”

(11) In preferred embodiments, edge service **112** operates a peer-to-peer service managing peer-to-peer connections among service devices **115**, illustrated herein in a preferred embodiment as wireless devices, thus creating additional paths of communication and enabling management platform **100** to optimize data transmission across multiple paths of communication, including paths across peer-to-peer links. This allows for more robust communication in many instances: For example, a service device **115** may have an unreliable network connection or lose a wireless signal, but if it maintains one or more peer-to-peer links with other devices **115**, terminal devices **116** can still access cloud service **110** and its managed applications and services **118**. Moreover, if a service device **115** has a strong signal or high data throughput, but is idle and not actively transferring data, management platform **100** can route the data transfer of a second service device **115** through the idle service device **115** in order to allow second service device **115** to benefit from the strong signal or high throughput of the first service device **115**.

(12) End-user interface **114** interacts directly with the user and cloud service **110** to deliver the user access to managed applications and services **118** that are managed by the cloud service **110**. Preferred embodiments of end-user interface **114** also act as the individual management interface for the various available applications and services, providing value-added functions such as single sign-on authentication and a consistent user experience.

(13) Preferred embodiments of methods to improve performance of wired and wireless networks utilize all three components, including cloud service **110**, edge service **112**, and end-user interface **114**, in order to fully realize the greatest optimization possible.

(14) Edge service **112** and end-user interface **114** can be combined, but preferred embodiments require physical and logical separation of edge service **112** and cloud service **110** for optimal functioning of a method to improve performance of wired and wireless networks. The separation allows for management of both ends of the data transmission process, as illustrated further in FIG. **3**, from end-user interface **114** when it is operating on the Internet or wide area network or is otherwise not able to communicate with edge service **112** directly. Even when only one end of the data transmission process is managed, as illustrated in FIG. **4**, a separation is preferable between edge service **112**, which runs across service devices **115** in preferred embodiments and on a local network to devices **115** in alternate preferred embodiments, from cloud service **110** which runs on the Internet or wide area network across which data is being transmitted. Furthermore, separation of edge service **112** from cloud service **110** allows for analytics and data management to be performed on more capable hardware, thus reducing the overall cost of edge service **112**. Additional This separation allows for service quality to be monitored and procedures such as data timing and error detection to be used in determining optimal paths for data transfer may also be employed by this separation.

(15) In use, management platform **100** is responsible for controlling all processes related to authentication and security, edge services, data management, data communications, device management, user authorization, applications services, and monitoring. Data management services provided by management platform **100** include management and optimization of all data throughput, communication path management, and quality of service (QoS). Management platform **100** is operable with third-party wired and/or wireless networks for conveying both broadband and data communications.

(16) In a preferred embodiment, devices **115**—including wired and/or wireless devices—with optimized antennas, CPU, memory, and additional wired and wireless communications capabilities

such as WiFi and Ethernet are used.

(17) Edge service **112** running on one or more service devices **115** manages network protocols and traffic and provides management and telemetry interface services with management platform **100**. In some embodiments, the application service managed by cloud service **110** and edge service **112** not only monitors performance and activity of terminal devices **116**, but also assigns and manages priority of terminal devices **116**, or of applications and services accessed by terminal devices **116**, or both. Thus, data traffic determined to be more important can be given priority.

(18) End-user interface **114** provides access to external services communicated across the above-described capabilities.

(19) Preferred embodiments of management platform **100** and method **200** (shown in FIG. **6**) are used by an end user as follows: The end user installs a wired and/or wireless service device **115**, such as mobile telephone **116**C, with the edge service **112** installed at their house or business. Upon powering up the service device **115**, the service device **115** connects to the cloud service **110**, authenticates, and registers with the cloud service **110**. Once connected and registered with cloud service **110**, a software download or update places the latest available software on the service device **115**. More particularly, once a connection to the cloud service **110** is available and the user registers, the user is prompted to download the software; for example, in a preferred embodiment, a user registering with a mobile telephone **116**C is provided with a link to download or update the software via an app store. In other cases, the software may be obtained by download or other means at any point prior to powering up device **115**. Once this software is installed on service device **115**, the user is able to manage the access for end-user terminal devices **116** (e.g., laptops, tablets, mobiles, peripherals) with the service device **115** including all networks and applications that they are authorized to access. More service devices **115** may be then registered using the same process to form a single, cohesive, multipath, fault tolerant edge service **112**. Once this is completed, the edge service **112** and cloud service **110** are ready to perform the optimization methods and processes to deliver an excellent user experience to the end-user on all terminal devices **116**. All control and interface with the end user are facilitated through this software.

(20) Referring now to FIG. **2**, in a preferred embodiment of management platform **100**, edge service **112** creates additional network paths in order to allow for improvement of network performance in cases in which all existing data paths for some devices **116** are suboptimal. Service devices **115** are illustrated herein in a preferred embodiment of wireless devices, but alternative embodiments in the form of wired service devices are fully contemplated. Moreover, a single service device **115** can have both wired and wireless networking capabilities, as illustrated by device **115**Z. It is fully contemplated that embodiments of platform **100** and method **200** (shown in FIG. **6**) can use any combination of types of service devices **115**.

(21) Service devices **115**, illustrated as wireless devices **115**W, **115**X, **115**Y, and **115**Z each have a communication path, illustrated here as a wireless communication path **126**, across network **124** to a Wide Area Network **130**, which in the most common use case will be the Internet; in some preferred embodiments, the connection at the Internet **130** end will be to cloud service **110** in particular, as illustrated in FIG. **3**. In exemplary cases, the communication path across network **124** involves a connection to a switch, router, wireless router, mobile tower, repeater, or other wired or wireless networking apparatus. Moreover, some service devices **115** may have multiple communication paths **126** or **128** (illustrated in FIGS. **4** and **5**), such as a computer with both a wired and wireless connection, or a tablet or mobile phone with both WiFi and mobile phone—e.g. Long-Term Evolution (LTE) or other mobile telecommunications standard—service.

(22) In some cases, a communication path may be obstructed or disconnected, weakening or entirely disabling communications along network **124**. For example, path **126** of **115**W is illustrated as a wireless signal by which service device **115**W is connected to Internet **130**, but which is interfered with by mountain **119**. Therefore, service device **115**W has a weak or nonexistent signal through which it accesses the Internet **130**, as shown by the dotted lines for its

communication path **126**, causing both speed and reliability problems.
(23) Edge service **112**, running on all service devices **115**W, **115**X, **115**Y, and **115**Z, has created peer-to-peer connections **122** between service device **115**W and service device **115**Y, as well as between service device **115**Y and service device **115**Z. Both service devices **115**Y and **115**Z have much more reliable communication paths **120** than service device **115**W. However, the peer-to-peer connections **122** create two virtual communication paths for service device **115**W to access the Internet **130**.
(24) More particularly, in addition to its obstructed communication path **126**, the peer-to-peer connection **122** from service device **115**W to terminal device **115**Y allows data to be transferred from service device **115**W to service device **115**Y, then over to the Internet **130** through the communication path **126** of terminal device **115**Y, and from Internet **130** to service device **115**W through the same path in reverse; this constitutes one alternate communication path for service device **115**W.
(25) Since service device **115**Y and service device **115**Z have a peer-to-peer connection **122** between them, this provides a second alternate communication path for service device **115**W: Data can be transferred from service device **115**W to service device **115**Y, then from service device **115**Y to service device **115**Z, and then to the Internet **130**. Responses can be sent back from a service on the Internet **130** to service device **115**W through the reverse path. Should communication path **126** of service device **115**Y become unavailable, both service device **115**W and service device **115**Y can continue to engage in data transfer through their direct or indirect connections to service device **115**Z. As a result, greater resiliency is provided to network **124**, providing greater reliability for communications. Moreover, throughput can be improved by transferring data across multiple paths, both direct communication paths across network **124** and the virtual paths created by peer-to-peer connections **122**, at the same time. This can be further facilitated by breaking up a message into parts and sending portions of the message across different paths to be reconstructed by cloud service **110**, as discussed below in connection with FIG. **3**.
(26) Referring now to FIGS. **3-5**, various implementations of management platform **100** for particular use cases are illustrated. Although these particular embodiments are discussed in detail for illustrative purposes, other embodiments involving the various possible combinations of components described below are fully contemplated.
(27) Referring now to FIG. **3**, a preferred embodiment of a system for improving performance of wireless and wired networks is illustrated and generally designated **101**. System **101** uses management platform **100** (see FIG. **1**) to actively manage both ends of the data transmission process between devices **116** and services on the Wide Area Network (WAN) or Internet **130** while maintaining full custody of the data before transmission over the Internet **130**.
(28) In use, terminal devices **116** communicate through edge service **112** over network **124** with cloud service **110**. Network **124** includes at least a wireless network or communication path **126** or a wired network or communication path **128**, and in some embodiments includes more than one, or one or more of each. In this configuration, terminal devices **116** access online applications or services on the Internet **130** through cloud service **110**. Therefore, between edge service **112** managing communications at the terminal device **116** end of the data transmission process, and cloud service **110** managing communications at the Internet or WAN **130** end, both ends of the communication process are actively managed.
(29) Edge service **112** monitors performance of each terminal device **116**. Telemetry **132** and timing data **134** is provided to cloud service **110** for analytics **136**, which involves comparing transfer speed and reliability across communication paths and determining optimal paths for data transfer. Telemetry data **132** may be provided as a single, combined, view of all terminal device data **116** across networks **124**. Preferred embodiments separate telemetry data **132** into groups using filters defined by cloud service **110**. Such a filter may describe data specific to a particular application or service **118**. Preferred embodiments define a set of curated filters that allow analytics

136 to refine the optimization of data flow across networks **124** for each terminal device **116**.

(30) Policy and data optimization control **138** is provided by cloud service **110**, based on which edge service **112** monitors and controls terminal devices **116**. Policy decisions and data flow optimizations **138** are provided as filter specifications **140**, which includes directions to edge service **112** to route data transfer through communication paths identified by analytics **136** as optimal for each terminal device **116**, as well as instructions for prioritizing access for specific terminal devices **116** or restricting access to certain services for specific terminal devices **116**.

(31) Multiplexing and inverse multiplexing technology is used to deconstruct and reconstruct communications messages for delivery across the multiple available data paths across network **124**, as is made possible by management of both ends of the data transmission process. This overcomes limitations of existing Internet routers today that prevent terminal devices **116** using multiple paths simultaneously to access the Internet **130**. Machine-to-machine data protocols are used to monitor, manage, and optimize the wireless communications of each wired and/or wireless terminal device **116** and the networks **124** or data paths across the networks **124**.

(32) Referring now to FIG. **4**, a preferred embodiment of a system for improving performance of wireless and wired networks is illustrated and generally designated **102**. System **102** uses management platform **100** (see FIG. **1**) to actively manage one end of the data transmission process while maintaining full custody of the data before transmission over the Internet **130**.

(33) In the illustrated embodiment, edge service **112** manages data transfer between terminal devices **116** across network **124** to the Internet or WAN **130**. Although cloud service **110** does not manage the transfer of data between network **124** and a service or application on the Internet **130**, it still receives telemetry data **132** and timing data **134** from edge service **112**, and provides policy **138** for edge service **112** to implement control over terminal devices **116**. This allows for analysis and selection of optimal data paths despite active management of only one end of the data transfer process.

(34) Referring back to FIG. **3**, multiplexing technology is used to manage the communications across the multiple available data paths through networks **124**. In both FIG. **3** and FIG. **4**, machine-to-machine data protocols are used to monitor, manage, and optimize the wireless communications of each wired and/or wireless terminal device **116** and the networks **124**. Preferred embodiments use highly efficient, lightweight, and fault tolerant IoT machine-to-machine data protocols for the analytics plane **142** and control plane **144**. One such industry standard that may carry both simultaneously is MQTT.

(35) Referring now to FIG. **5**, a preferred embodiment of a system for improving performance of wireless and wired networks is illustrated and generally designated **103**. System **103** uses management platform **100** (see FIG. **1**) to actively manage both ends of the data transmission process over a wireless data connection.

(36) Data service APIs **160** and application service APIs **162** are provided by cloud service **110** and edge service **112** in order to allow effective monitoring and control of terminal devices **116**, thus allowing management platform **100** to calculate optimal data transfer paths and manage data distribution across available paths, in view of priority of terminal devices **116**, applications, and services, when priorities are assigned, as discussed below. The application service or services support the management of terminal devices **116**, while the data management service or services perform the multipath data optimization.

(37) A preferred embodiment of the application service executes across all devices **115**, which have independent computer processing capability.

(38) In a preferred embodiment, the application service manages communication and user-based security between the terminal devices **116** and peripheral devices, such as scanners or printers, utilizing the data service. The application service also monitors performance and activity of terminal devices **116**, applications and services utilizing the data service. Moreover, in a preferred embodiment, a user of management platform **100** (see FIG. **1**) can prioritize data transfer for

specific terminal devices **116**, applications and services on the Internet **130**, or both, so that management platform provides preference to prioritized devices **116** and applications and services in assigning the most efficient communication paths when multiple devices **116** are engaged in data transfer. In accordance with the user-provided settings, the application service assigns and manages priority of terminal devices **116**, applications and services utilizing the data service.
(39) The edge service **112** and cloud services **110** managing the data transmission process provide the ability for the Data Service APIs **160** and the Application Service APIs **162** to effectively function across platform **100** (see FIG. **1**) and method **200** (See FIG. **6**) implemented on platform **100**. The methods used to optimize data conductivity between the Cloud Service **110** and the Edge Service **112** utilize existing and novel algorithms to allow the conducting of data across multiple wired and wireless devices **116** to create a fault-tolerant, high-performance connection through the multi-device collaboration of services **164**. In a preferred embodiment, such algorithms may be rules-based, goal-based, or a combination of both. Communication between the devices **115** to enable this edge service **112** is via peer-to-peer data communications via wired and/or wireless connectivity between the devices **115**, as described above in connection with FIG. **2**. These edge services **112** are managed and monitored over a direct data management interface **166** separate from the data conveyance layer **168** (shown in dashed lines) that creates and maintains a connection between the edge service **112** and the cloud service **110**, providing a higher level of data visibility and data management capability.
(40) Referring now to FIG. **6**, a preferred embodiment of a method for improving performance of wireless and wired networks is illustrated and generally designated **200**. As illustrated, method **200** begins with step **210** of providing one or more devices **115** (see FIG. **1**) configured for communication—data transfer in a preferred embodiment, but it is also contemplated that the method can be applied to other forms of telecommunication, such as digital or analog voice—across a network. In preferred embodiments, the network includes one or more wired networks, or one or more wireless networks, or one or more of each. One or more terminal devices **116** (see FIG. **1**) that use devices **115** to access applications or services **118** (see FIG. **1**) are also provided.
(41) Step **210** is not limited to a single moment or operation: It is contemplated that in many situations additional devices **115** will at times become available for multi-device collaboration (such as is illustrated in step **218** below, for example), and on occasion a device **115** may become unavailable, for example by signal loss.
(42) Preferred embodiments of method **200** include step **214**, which involves providing a management platform **100** (shown in FIG. **1**) configured for monitoring and managing communications over the network **124** (shown in FIGS. **2-4**). Management platform **100** can be configured according to the embodiments illustrated in FIGS. **2-4**, or another embodiment including any combination of features and elements disclosed herein. Portions of some embodiments of management platform **100**, such as some embodiments of edge service **112**, are installed on devices **115** and act in concert to operate as if a single process across multiple devices **115**. In such cases, management platform **100** is capable of being continuously modified in response to changes in network **124**, such as the addition of a new device **115** or loss of communication with a device **115**. In preferred embodiments, management platform **100** operates to perform the following steps.
(43) When multiple devices **115** are provided, preferred embodiments of method **200** include step **218** of creating peer-to-peer communication paths **122** (shown in FIG. **2**) between the available devices **115**. As a result of performing step **218**, additional communication paths are created beyond the existing (e.g., third party) wired and wireless networks **124** (shown in FIGS. **2-4**) across which data transfer is being or planned to be performed. As with step **210**, step **218** can be repeated in response to newly available devices **115** or the loss of communication with a device **115**.
(44) Step **222** involves the monitoring of service quality across each path of communication. In preferred embodiments, step **222** is performed continuously, or at least repeatedly, in order to

respond to changes in service quality across different paths of communication.
(45) In step **226**, the optimal path of communication between a terminal device **116** and the wide area network or Internet **130** is determined. In preferred embodiments, step **226** is performed using one or more of data timing, error detection, and active bandwidth detection. An exemplary embodiment treats resolution of this data into routing and prioritization controls as a form of vehicle routing problem and use methods such as shortest-path or quickest-path to optimize traffic over networks **124**. Moreover, in preferred embodiments, step **226** is performed for each terminal device as it accesses applications or services **118** (see FIG. **1**), and is performed continuously or repeatedly in order to allow step **230** to provide optimal network performance.
(46) In step **230**, the results of steps **222** and **226** are used to manage data distribution in the most efficient manner possible across all available paths of communication. As with the other steps, step **230** is performed continuously or repeatedly, as shown by repeating loop path **234**, in order to respond to changing network conditions, allowing for optimum performance, and more particularly, enhanced overall quality, throughput speed, and reliability of data transfer at all times.
(47) While there have been shown what are presently considered to be preferred embodiments of the present invention, it will be apparent to those skilled in the art that various changes and modifications can be made herein without departing from the scope and spirit of the invention.

Claims

1. A system for improving network performance, comprising: a management platform comprising a cloud service and an edge service; one or more service devices operating the edge service; and one or more user devices configured for accessing online applications and services through the one or more service devices, wherein the edge service comprises traffic and network monitors configured to provide telemetry and timing data to the cloud service, and the cloud service uses the telemetry and timing data to determine optimal paths and method of transmission across the Internet through the one or more service devices and provide policy to the one or more service devices, thereby causing the one or more service devices to utilize all available paths of communication between the one or more service devices to optimize data throughput and quality; and responding to a network disruption affecting a first service device of the plurality of service devices by transferring data between the first service device and a second service device of the plurality of service devices, the second service device being unaffected by the network disruption.
2. The system for improving network performance as recited in claim 1, wherein the one or more service devices comprises at least two user devices.
3. The system for improving network performance as recited in claim 2, wherein peer-to-peer data communications are utilized between the service devices in order to create additional data transmission paths.
4. The system for improving network performance as recited in claim 2, wherein the edge service comprises shared computing service created across the service devices to manage the optimal paths of data.
5. The system for improving network performance as recited in claim 1, further comprising an application service configured to support the management of the one or more user devices.
6. The system for improving network performance as recited in claim 5, further comprising a data service, wherein the application service manages the communication between the one or more user devices and peripheral devices configured to utilize the data service.
7. The system for improving network performance as recited in claim 5, wherein the application service is configured to manage user-based security between the one or more user devices.
8. The system for improving network performance as recited in claim 5, wherein the application service is configured to monitor performance and activity of the one or more user devices.
9. The system for improving network performance as recited in claim 5, wherein the application

service is configured to assign and manage priority of the one or more user devices.

10. A method for improving network performance, comprising the steps of: providing a plurality of service devices configured for communicating data over a network; providing a cloud service configured to manage and direct the plurality of service devices; providing one or more terminal devices configured for accessing applications and services over a network using the plurality of service devices; sending telemetry and timing data from the plurality of service devices to the cloud service; determining, using the telemetry and timing data from the plurality of service devices, the optimal path of communication across the Internet through a service device of the plurality of service devices for each of the one or more terminal devices; establishing policy for the plurality of service devices, causing the one or more service devices to use the optimal path of communication for each of the one or more terminal devices; and responding to a network disruption affecting a first service device of the plurality of service devices by transferring data between the first service device and a second service device of the plurality of service devices, the second service device being unaffected by the network disruption.

11. The method for improving network performance as recited in claim 10, further comprising the step of providing a management platform configured for monitoring and managing communications, wherein the management platform performs the steps of monitoring service quality, determining the optimal path of communication, and managing data distribution across the available paths of communication.

12. The method for improving network performance as recited in claim 10, wherein the one or more service devices comprise at least two service devices.

13. The method for improving network performance as recited in claim 12, further comprising the step of creating peer-to-peer communication paths between the service devices in order to provide additional communication paths for the steps of monitoring service quality, determining the optimal path of communication, and managing data distribution across the available paths of communication.

14. The method for improving network performance as recited in claim 12, further comprising the step of creating a shared computing service across the at least two service devices that manages optimal paths of data.

15. The method for improving network performance as recited in claim 12, wherein the steps of monitoring service quality, determining the optimal path of communication, and managing data distribution across the available paths of communication use machine-to-machine data protocols to monitor, manage, and optimize multipath communications with the terminal devices.

16. The method for improving network performance as recited in claim 10, wherein the step of managing data distribution across available paths of communication comprises the use of multiplexing to manage communication across multiple paths.

17. The method for improving network performance as recited in claim 10, wherein the steps of monitoring service quality, determining the optimal path of communication, and managing data distribution across the available paths of communication use machine-to-machine data protocols to monitor, manage, and optimize multipath communications between the one or more terminal devices and the network.

18. The method for improving network performance as recited in claim 10, wherein the network comprises a wired network and one or more wireless networks.

19. The method for improving network performance as recited in claim 10, wherein the step of determining the optimal path of communication is performed using data timing and error detection.

20. The method for improving network performance as recited in claim 10, wherein the step of determining the optimal path of communication is performed using active bandwidth detection.